

DIGITAL SUMS OF POWERS, FACTORIALS, AND LEAST COMMON MULTIPLES

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ABSTRACT. We prove logarithmic lower bounds for digital sums of powers, multiples of powers, factorials, and least common multiples, using only elementary number theory. We then explain how Stewart's theorem, via lower bounds for linear forms in logarithms, gives the corresponding general result for powers.

1. INTRODUCTION

In this article, we prove logarithmic lower bounds on digital sums of powers, multiples of powers, factorials, and the least common multiple of $\{1, \dots, n\}$, using only elementary number theory. We then explain how Stewart's theorem, via lower bounds for linear forms in logarithms, establishes a sublogarithmic lower bound on the number of nonzero base- b digits of a^n when $a, b \geq 2$ are multiplicatively independent.

Conjecturally, the sum of the base- b digits of a^n should grow linearly in n for multiplicatively independent a and b , but this remains unproved.

2. A DIVISIBILITY ARGUMENT FOR POWERS

Let $s_b(N)$ denote the sum of the base- b digits of N , and let $c_b(N)$ denote the number of nonzero base- b digits. Since

$$c_b(N) \leq s_b(N) \leq (b-1)c_b(N),$$

the functions c_b and s_b are equivalent up to constants depending only on b .

The following simple lemma is the key idea. Throughout, $a, b \geq 2$ are integers.

Lemma 1. *Let $2 \leq a < b$ with $a \mid b$. Let $(e_k)_{k \geq 1}$ be integers such that $e_1 \geq 1$ and $a^{e_k} > b^{e_{k-1}}$ for $k \geq 2$. If N is divisible by a^{e_k} but not by b , then $c_b(N) \geq k$.*

Proof. The proof is by induction on k . For $k = 1$ the claim is immediate. For $k \geq 2$, write

$$N = b^{e_{k-1}}q + r, \quad 0 \leq r < b^{e_{k-1}}.$$

Since $N \geq a^{e_k} > b^{e_{k-1}}$, we have $q \geq 1$. Also r is divisible by $a^{e_{k-1}}$, because both N and $b^{e_{k-1}}q$ are; and r is not divisible by b , because N is not. Hence, by induction, $c_b(r) \geq k-1$. The base- b expansion of N is obtained by adjoining the block of e_{k-1} low-order digits represented by r to the expansion of q , so

$$c_b(N) = c_b(q) + c_b(r) \geq 1 + (k-1) = k.$$

□

Choosing $e_1 = 1$ and $e_k = \lceil re_{k-1} \rceil$, where $r = \log b / \log a > 1$, gives $e_k < r^k / (r-1)$. Lemma 1 therefore implies the following explicit form.

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Corollary 1. *Let $2 \leq a < b$ with $a \mid b$. If N is divisible by a^n but not by b , then*

$$c_b(N) > C \log n$$

for all sufficiently large n , where $C > 0$ depends only on a and b . Moreover, any $C < 1/\log(\log b / \log a)$ is admissible.

To apply this to powers where b divides a^n , one must remove the trailing zeros in base b . The next lemma ensures that, under a multiplicative independence hypothesis, some prime survives to high order after those trailing zeros are removed.

Lemma 2. *Let $a, b \geq 2$, and suppose $\log a / \log b$ is irrational. If*

$$a^n = b^m t,$$

with $t \in \mathbb{Z}$, then there is a prime $p \mid a$ and a constant $C > 0$, depending only on a and b , such that

$$\nu_p(t) \geq Cn,$$

where $\nu_p(t)$ denotes the p -adic valuation of t .

Proof. Since $\log(a)/\log(b)$ is irrational, there are no integers u, v with $a^v = b^u$, except $u = v = 0$. Equivalently, the vectors $(\nu_p(a))_p$ and $(\nu_p(b))_p$ are linearly independent over $\mathbb{Q}$. Hence there are primes p, q with

$$\nu_p(a)\nu_q(b) - \nu_q(a)\nu_p(b) > 0.$$

Comparing p -adic and q -adic valuations in $a^n = b^m t$ gives

$$n\nu_p(a) = m\nu_p(b) + \nu_p(t), \quad n\nu_q(a) \geq m\nu_q(b).$$

Eliminating m yields

$$\nu_p(t) \geq n \left(\nu_p(a) - \frac{\nu_q(a)}{\nu_q(b)} \nu_p(b) \right),$$

and the coefficient is positive by the choice of p, q . □

Corollary 2. *If the conditions of Lemma 2 hold then $\log t \asymp_{a,b} n$.*

Proof. The upper bound follows from $t \leq a^n$. The lower bound follows from Lemma 2 because t is divisible by $p^{\lceil Cn \rceil}$, hence $\log t \geq Cn \log p$. □

Let d be the largest divisor of a all of whose prime factors divide b . Equivalently, d is the smallest divisor of a such that a/d is coprime to b . Combining Corollary 1 with Lemma 2 gives a logarithmic lower bound when d and b are multiplicatively independent.

Theorem 1. *Let $a, b \geq 2$, let d be the smallest divisor of a such that $\gcd(a/d, b) = 1$, and suppose that $\log d / \log b$ is irrational. Then*

$$c_b(a^n) > C \log n$$

for all sufficiently large n , with $C > 0$ depending only on a and b .

Proof. Let $g = a/d$, and write $a^n = b^m s$ with $b \nmid s$. Then g^n divides s , since g^n divides a^n and it is coprime to b^m , so let $t = s/g^n$. Then $d^n = b^m t$. By Lemma 2, there is a prime $p \mid d$ with $\nu_p(t) \geq C_1 n$. Since $t \mid s$, we also have $\nu_p(s) \geq C_1 n$. Applying Corollary 1 to s gives $c_b(s) > C \log n$. Finally, a^n and s differ only by a power of b , so they have the same nonzero base- b digits. □

In particular, $c_{10}(a^n)$ grows at least logarithmically in n if the prime factorization of a contains unequal numbers of twos and fives. We encounter an obstacle when attempting to generalize this result to other powers, because Lemma 1 applies to multiples of a^n , and these multiples may have few nonzero digits. For example, every 3^n has a multiple of the form $10^k + 8$ with only two nonzero decimal digits. Thus we need different techniques to prove Stewart's theorem in full.

3. FACTORIALS AND LEAST COMMON MULTIPLES

For factorials and least common multiples, the useful divisibility is not by a large prime power but by numbers of the form $b^r - 1$. The following lemma was first proved by Stolarsky [7, Theorem 2.1] for base 2, and extended to general bases by Balog and Dartyge [1, Lemma 4].

Lemma 3. *If $m, r \geq 1$ and $b \geq 2$, and if $b^r - 1 \mid m$, then*

$$s_b(m) \geq (b - 1)r.$$

Proof. Group the base- b expansion of m into blocks of length r :

$$m = \sum_i B_i b^{ri}, \quad 0 \leq B_i < b^r.$$

Let $G(m) = \sum_i B_i$. Since $b^r \equiv 1 \pmod{b^r - 1}$, we have $G(m) \equiv m \pmod{b^r - 1}$. Moreover $G(m) < m$ whenever $m \geq b^r$, while $G(m) = m$ for $m < b^r$. Iterating G therefore carries any positive multiple of $b^r - 1$ down to the unique such multiple below b^r , namely $b^r - 1$. Subadditivity of digital sums gives

$$s_b(G(m)) \leq \sum_i s_b(B_i) = s_b(m),$$

so $s_b(m) \geq s_b(b^r - 1) = (b - 1)r$. □

Taking $m = n!$ or $m = \Lambda_n = \text{lcm}(1, \dots, n)$, we obtain a logarithmic lower bound immediately. If $n \geq b^r - 1$, then $b^r - 1$ divides both $n!$ and Λ_n . With $r = \lfloor \log_b(n + 1) \rfloor$, Lemma 3 yields

$$s_b(n!) \geq (b - 1)\lfloor \log_b(n + 1) \rfloor, \quad s_b(\Lambda_n) \geq (b - 1)\lfloor \log_b(n + 1) \rfloor.$$

Thus both digital sums are $\gg_b \log n$. Similar arguments appear in work of Luca [3]; Sanna [4], building on this lemma and deeper methods, proved the stronger bound

$$s_b(n!) > C_b \log n \log \log n$$

for $n > e^e$, and the same estimate holds for $s_b(\Lambda_n)$.

Heuristically, after ignoring trailing zeros, the remaining digits of $n!$ and Λ_n should be roughly equidistributed among $\{0, 1, \dots, b - 1\}$. This suggests much larger orders of growth, namely $s_b(n!) \asymp n \log n$ and $s_b(\Lambda_n) \asymp n$, but such estimates are presently out of reach.

4. STEWART'S THEOREM

Senge and Straus [5, Theorem 3] showed that for integers $a, b \geq 2$,

$$\lim_{n \rightarrow \infty} c_b(a^n) = \infty \quad \text{if and only if} \quad \frac{\log a}{\log b} \text{ is irrational.}$$

However, their result does not yield any explicit lower bound. The irrationality hypothesis is necessary because if $\log a / \log b = r/s$ for positive integers r and s , then $a^{sn} = b^{rn}$ has only one nonzero base- b digit for every n .

Subsequently, Stewart [6, Theorem 2] proved the following effectively computable lower bound.

Theorem 2. *Let $a, b \geq 2$, and suppose $\log a / \log b$ is irrational. Then there is a constant C , depending only on a and b , such that*

$$c_b(a^n) > \frac{\log n}{\log \log n + C} - 1$$

for all $n > 4$.

Proof. Let $N = a^n$, and write its nonzero base- b digits in the form

$$N = b^m (d_1 b^{-m_1} + \cdots + d_k b^{-m_k}),$$

where

$$1 = m_1 < m_2 < \cdots < m_k \leq m, \quad 1 \leq d_j \leq b-1.$$

Here m_j measures the distance of the j -th nonzero digit from the leading position: the j -th nonzero digit occurs at exponent $m - m_j$.

We shall prove that consecutive nonzero digit positions cannot be separated by too large a multiplicative factor. Fix $i < k$. Split N at the i -th nonzero digit:

$$q = b^{m_i} \sum_{j \leq i} d_j b^{-m_j}, \quad r = b^m \sum_{j > i} d_j b^{-m_j}.$$

Then q and r are positive integers and

$$N = b^{m-m_i} q + r.$$

Moreover,

$$b^{m_i-1} \leq q < b^{m_i},$$

because q is the integer formed by the first m_i digits of N , and its leading digit is nonzero. Similarly, since the first nonzero digit of r occurs at exponent $m - m_{i+1}$, we have

$$b^{m-m_{i+1}} \leq r < b^{m-m_{i+1}+1}.$$

Since $b^{m-1} < N < b^m$, it follows that

$$b^{-m_{i+1}} < \frac{r}{N} < b^{-m_{i+1}+2}.$$

Now define the linear form

$$\Lambda = -n \log a + (m - m_i) \log b + \log q.$$

This quantity is nonzero. Indeed, if $\Lambda = 0$, then $N = b^{m-m_i} q$, which would force $r = 0$, contradicting the fact that there is at least one nonzero digit after the i -th one.

Applying a theorem of Baker and Wüstholz [2, p. 20] to this form yields the lower bound

$$\log |\Lambda| > -C_1 \max\{1, \log q\} \log n.$$

If $m_i \geq 3$, then $q \geq b^{m_i-1} \geq b^2$, so in particular $\log q > 1$, and the preceding estimate becomes

$$\log |\Lambda| > -C_1 \log q \log n.$$

Using $q < b^{m_i}$, this gives

$$\log |\Lambda| > -C_2 m_i \log n.$$

On the other hand,

$$e^\Lambda = 1 - \frac{r}{N}.$$

Since $q \geq 1$ and $r < b^{m-m_i}$, the decomposition $N = b^{m-m_i}q + r$ gives $N > 2r$. Thus $r/N < 1/2$, and so

$$|\Lambda| = -\log(1 - r/N) < 2r/N.$$

Using $r/N < b^{-m_{i+1}+2}$, we obtain

$$\log |\Lambda| < \log 2 + (-m_{i+1} + 2) \log b.$$

Combining the lower and upper bounds for $\log |\Lambda|$ yields

$$-C_2 m_i \log n < \log 2 + (-m_{i+1} + 2) \log b,$$

and hence

$$m_{i+1} \leq C_3 m_i \log n.$$

This proves the desired estimate when $m_i \geq 3$.

If $m_i < 3$, then q is bounded in terms of b . In this case the Baker–Wüstholz lower bound gives simply

$$\log |\Lambda| > -C_4 \log n.$$

The same upper bound

$$\log |\Lambda| < \log 2 + (-m_{i+1} + 2) \log b$$

then implies

$$m_{i+1} \leq C_5 \log n.$$

Since $m_i \geq 1$, this again gives

$$m_{i+1} \leq C_6 m_i \log n.$$

Thus for every $i < k$,

$$\frac{m_{i+1}}{m_i} \leq C_6 \log n.$$

Taking logarithms gives

$$\log m_{i+1} - \log m_i \leq \log \log n + C_7.$$

Summing over $i = 1, \dots, k-1$, and using $m_1 = 1$, we get

$$\log m_k \leq (k-1)(\log \log n + C_7).$$

It remains to relate m_k to n . Let T be the number of trailing zeroes in the base- b expansion of $N = a^n$, and write

$$N = b^T t, \quad b \nmid t.$$

The last nonzero digit of N occurs at exponent T . Since that same digit has exponent $m - m_k$, we have

$$T = m - m_k.$$

Therefore $t = N/b^T$ has exactly m_k base- b digits. Equivalently,

$$b^{m_k-1} \leq t < b^{m_k},$$

so

$$m_k = \frac{\log t}{\log b} + O(1).$$

By Corollary 2,

$$\log t \asymp_{a,b} n.$$

Hence

$$m_k \asymp_{a,b} n,$$

and therefore

$$\log m_k = \log n + O_{a,b}(1).$$

Combining this with the previous inequality gives

$$\log n + O_{a,b}(1) \leq (k-1)(\log \log n + C_7).$$

After increasing the constant in the denominator, we obtain

$$k > \frac{\log n}{\log \log n + C} - 1$$

for all $n > 4$. Since $k = c_b(a^n)$, this proves the theorem. $\square$

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